

Decision-making System based on Finite State Machine for Low-speed Autonomous Vehicles in the Park

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Abstract— Park is an important population and industry gathering area in a city, and a popular scene in theoretical research and technical application of automatic driving. Decision-making, path planning and control are the core technologies. Due to cost and safety issues, automatic driving has not been widely used for park applications. In this paper, a decision-making model is devised based on finite state machine (FSM), a local path planner is designed based on Bézier curve and the lateral and longitudinal control method is realized on GAC Aion LX. This decision-making system for low speed automatic driving in the park is tested. Results demonstrate its accuracy and reliability.

I. INTRODUCTION

As an important scene to realize autonomous driving, low-speed autonomous vehicles in the park have great commercial application potential. Decision-making, path planning and control system are the key technologies of automatic driving, in which much research has been performed.

Finite state machine (FSM) is widely used in decision-making system. Boss's FSM based decision-making system [1] defines the highest level based on driving scenes, and defines the bottom layer based on driving behaviors. Three highest levels and their corresponding bottom layers are lane driving (made up by lane selector, merge planner, current scene reporter and distance keeper), intersection handling (made up by precedence estimator, pan-head planner and transition manager), and achieving a zone pose (made up by state estimator and goal selector) respectively. Talos [3], which is good at logical clarity, divides its decision-making system into lane finding, navigator, motion planner, controller, drivability map and obstacle detector by logical hierarchy. But it also has poor adaptability to complex problems. Junior [4] divides its decision system into lane changes, U-turns, parking, and merging into moving traffic. AnnieWAY [5] which includes all relevant maneuvers in DARPA Urban Challenge divides its decision system into DriveStart, DriveOnLane, LaneChange, DriveKTurn, DriveStop and DriveRecover. Team Cornell's [6] intelligent planning system uses the scene

estimator's probabilistic perception of the environment to plan mission paths within the context of the rule-based road network. However, there are still conditions that are not covered by their FSM in real test. Other scene-based FSM [7-8] design their decision making system aiming at urban environments and highway and verified their system through the simulation respectively.

This paper presents a new decision-making system with path planning and control methods for application in the park. The lane change path is manipulated by setting control points based on Bézier curve. A lateral and longitudinal control method is used for path tracking. The road test results of the decision-making algorithm demonstrate its accuracy and reliability.

This paper is organized as follows: Section II summarizes the system architecture. Section III describes the design of the decision-making system which includes planning and control method. Section IV shows the experiments and results of path tracking and lane changing. Section V provides the conclusions and discusses the future work.

II. ARCHITECTURE

Figure 1 shows the overall system architecture of this paper including the following subsystems.

Road-Point Detection using image-processing method to fit splines to lane based on cameras data provides lane boundry for drivability area.

Obstacles Detector uses lidar and cameras data to detect the position, velocity, moving direction, size and category of the objects.

Fusion of Positioning reconciles Inertial measurement unit (IMU) information, global positioning system (GPS) and Real-time kinematic (RTK) integrated positioning information and point cloud matching positioning information, and outputs the real-time positioning results with high precision finally.

The high precision map uses lidar and positioning data to mark the pre-defined path and the hazard which is hard to detect by Obstacles Detector, like hollow on the road.

Drivability Area reconciles the lane detection and high precision map, answering queries from Local Path Planner of the validity of drivability path.

The Path Tracker tracks the pre-defined path based on high precision positioning.

These modules are all running based on Robot Operating System (ROS), which provides efficient communication, a set of software libraries and tools.

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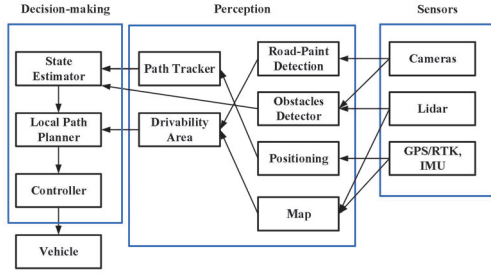


Figure 1. System architecture.

III. DECISION-MAKING SYSTEM BASED ON FSM

The FSM that is used to switch between different driving states is shown in Figure 2. There are 7 individual states in the FSM, which correspond to the following conditions.

StateEstimator: this state receives the vehicle's position, lane information, obstacles information and drivability area, and filters out the useless information.

StopAndWait: this state enables the autonomous vehicle to wait for pedestrians and other cars in the opposite lane passing when lane changing.

DistanceKeeping: information provided by perception system is used in this state to determine the necessary safety gaps between two vehicles and adjust the speed of the autonomous vehicle accordingly.

ConstantSpeed: in this state, the autonomous vehicle goes at the default speed which is 3m/s.

LaneChoosing: this state uses the perceptual information to determine the optimal lane to be in at any instant and evaluate whether it is feasible.

LocalPlanner: path planning based on Bézier curve is used in this state to merge into a lane proposed by lane choosing.

VehicleController: control system based on lateral and longitudinal control is used for path tracking.

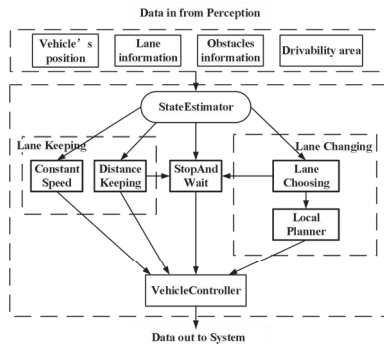


Figure 2. Behaviors architecture based on FSM

A. Distance Keeping

This behavior aims to make the actual distance not less than the desired distance between the autonomous vehicle and the vehicle in front. The following area is divided into Collision Avoidance Area and Vehicle Following Area according to d_{actual} , which means the actual distance between the front

vehicle and autonomous vehicle. For Vehicle Following Area, this means:

$$d_{stop} < d_{actual} < d_{stop} + d_{follow} \quad (1)$$

For Collision Avoidance Area, this means:

$$d_{actual} < d_{stop} \quad (2)$$

In Vehicle Following Area, the autonomous vehicle governs the speed according to the distance change in the past second: if d_{actual} decreases, then $v_d = v_d + 1$ ($v_d < 5 \text{ m/s}$); if d_{actual} increases, then $v_d = v_d - 1$ ($v_d > 0$); if d_{actual} is unchanged, then $v_d = v_d$, where v_d is the desired velocity.

In Collision Avoidance Area, $v_d = 0$.

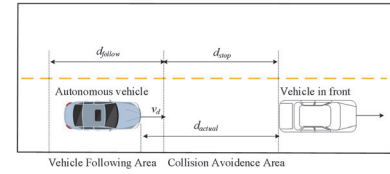


Figure 3. Distance Keeping state

B. Lane Choosing

In this paper, overtaking other cars on the road isn't considered. Hence, Lane Changing is only considered when avoiding static obstacles. Firstly, consider whether there are incoming vehicles in the opposite lane when encountering obstacles. If there are incoming vehicles in the opposite lane, stop and wait until the vehicles pass. Secondly, return to the original lane if passable, for staying in the opposite lane may cause accident.

C. Local Planner

The following three problems are one of the most difficult problems in path planning in order to reduce the probability of traffic accidents in the park[9].

(1) The path should meet the constraints of vehicle dynamics.

(2) The path should ensure driving comfort and safety.

(3) The path planner should be able to do path planning adaptively under different circumstances.

To solve the problems above, a real-time local path planner based on Bézier curve is proposed.

Nowadays, Bézier curves are widely used in computer graphics and animation, for which it has the advantages of continuous curvature, easily tracking and meeting vehicle dynamics constraints. Therefore, Bézier curve is selected for path planning in this paper.

A Bézier curve of degree n can be represented as

$$B(t) = \sum_{i=0}^n \binom{n}{i} b_{i,n}(t) P_i \quad (3)$$

where P_i are control points, $t \in [0, 1]$, $b_{i,n}(t)$ is a Bernstein polynomial given by

$$b_{i,n} = \binom{n}{i} t^i (1-t)^{n-i} \quad (4)$$

There are important properties of Bézier curve for path planning[10]:

- (1) They always pass through P_0 and P_n , and don't pass through other control points.
- (2) They are always tangent to the lines connecting $P_0 \rightarrow P_1$ and $P_{n-1} \rightarrow P_n$ at P_0 and P_n respectively.
- (3) They always lie within the convex hull consisting of their control points. This property can be used to constrain the position of control points, so as to ensure that the curve has no collision with obstacles.

Generally, the controller tracks the pre-defined path until coming across obstacles. This paper proposes a local planner to deal with lane changing in the park based on a fixed tracking path, which needs to follow the procedure as follows.

(1) Input parameters

The local planner requires parameters as follow.

v_{C_0} : the velocity of the obstacle, it's considered statics if $v_{C_0} < 0.2$, then local path will be planned. Otherwise, StateEstimator will turn to Lane Keeping state.

P_0 and P_5 : starting point $P_0 (X_0, Y_0)$ and ending point $P_5 (X_5, Y_5)$ are required for lane changing, where $P_5 (X_5, Y_5)$ is determined by High precision map and the position of obstacles, and $P_0 (X_0, Y_0)$ is where the vehicle is when obstacle is detected.

r : the safety distance between the autonomous vehicle and obstacles.

L_{C_0} and w_{C_0} : the length and width of the obstacle C_0 , which is provided by Perception system.

w_{C_1} : the width of the vehicle.

w_r : the width of the road which is provided by Road-Paint Detection.

$C_0(X_{ob}, Y_{ob})$: the center coordinate of the obstacles, which is provided by Perception system. If $X_{ob} > 12$, Local planner will ignore this object, for which it's still far away. If $8 \leq X_{ob} \leq 12$, Local planner will plan path to change lane, and the path will meet the vehicle dynamics constraints due to long distance for lane change. However, the behavior will turn to StopAndWait if $X_{ob} < 8$, because the path won't meet the vehicle dynamics constraints.

(2) Drivability area

The local planner in this paper treats vehicle as a point, which can result in collision due to the dimension of the vehicle. Hence, obstacle expansion radius r is used to avoid collision by increasing the size of the obstacle. Besides, d_{min} is

used for the minimum desired distance from obstacles to the vehicle. L is the desired distance between the vehicle and the obstacle, which can be written as

$$L = X_{ob} - d_{min} - \frac{L_{C_0}}{2} \quad (5)$$

According to property (3), the planning path is wrapped in a convex hull connected by its control points, so that it is only necessary to ensure that all control points of Bézier curve are limited in the safe area, then the curve will not collide with obstacles. Obviously, point O is the most prone to collision during lane changing as shown in Figure 4. This means the radius of circle O needs to satisfy:

$$r > w_{C_1} \quad (6)$$

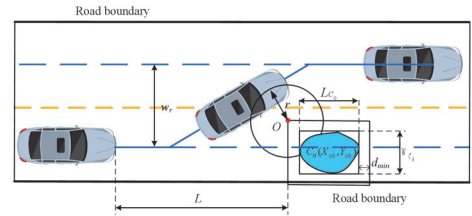


Figure 4. Lane change for obstacle avoidance

(3) Selection of control points

As shown in Figure 5, Bézier curve is used for local planning, where $P_i = (X_i, Y_i)$ are selected as the control points. This paper employs 5th-order Bézier curve for local planning. Point P_0 is the origin of local planning coordinates. Point P_5 is the end of lane change path, which is on pre-defined path 1, so that Y_5 can be given by the ordinate corresponding to X_5 . Take Figure 5 as an example, P_0 and P_5 are selected as

$$\begin{aligned} X_0 &= 0, & Y_0 &= 0 \\ X_5 &= L, & Y_5 &= w_r \end{aligned} \quad (7)$$

Divide L into four equal parts, after that P_1, P_2, P_3 and P_4 are computed by

$$\begin{aligned} X_1 &= \frac{X_5}{4}, & Y_1 &= Y_0, \\ X_2 &= \frac{X_5}{2}, & Y_2 &= Y_0, \\ X_3 &= \frac{X_5}{2}, & Y_3 &= Y_5, \\ X_4 &= \frac{3X_5}{4}, & Y_4 &= Y_5 \end{aligned} \quad (8)$$

After changing lane successfully, the vehicle needs to move past the obstacle. The distance that vehicle has to go on Pre-defined path 1 is decided by L_{line} , which is

$$L_{line} = L_{C_0} + 2d_{min} \quad (9)$$

Thus, the local planning has been constructed by 5th-order Bézier curve based on the 6 control points.

$$\kappa = \frac{2 \sin \alpha}{l_d} \quad (18)$$

where κ is the curvature of the arc, and the vehicle wheelbase is represented by L , then the expression of the rotation angle of the front wheel δ is

$$\delta = \tan^{-1}(\kappa L) \quad (19)$$

Combining (18) and (19), the control quantity of pure pursuit method can be written as

$$\delta(t) = \tan^{-1}\left(\frac{2L \sin(\alpha(t))}{l_d}\right) \quad (20)$$

A new variable e_1 is defined to represent the lateral error between the heading vector and the goal point. According to Ackermann model

$$\sin \alpha = \frac{e_1}{l_d} \quad (21)$$

The curvature of the circular arc can be written as

$$\kappa = \frac{2}{l_d^2} e_1 \quad (22)$$

The look-ahead distance l_d is expressed as a linear function of the vehicle longitudinal speed, which is $l_d = kv_x + l_0$, where v_x is the velocity of the vehicle, k is a coefficient and l_0 is the shortest look-ahead distance. The equation of the angle of front wheel results in

$$\delta(t) = \tan^{-1}\left(\frac{2L \sin(\alpha(t))}{kv_x(t) + l_0}\right) \quad (23)$$

Equation (23) is the control quantity of the pure pursuit method.

(2) Longitudinal control based on PID method

A PID method is used for longitudinal control, which is written as

$$U_t = K_p e(t) + \frac{1}{K_I} \int_0^t e(t) dt + K_D \frac{de(t)}{dt} \quad (24)$$

where U_t is the acceleration or braking control quantity of the vehicle; $e(t) = v_d - v_a$; v_d is the desired velocity while v_a is the actual velocity; and K_p, K_I, K_D are input parameters[11].

By discretizing (24), the calculation equation of discrete PID can be written as

$$U_t = K_p e_k + \frac{T}{K_I} \sum_{j=0}^k e_j + K_D \frac{e_k - e_{k-1}}{T} \quad (25)$$

where T is the instruction cycle of the controller, and U_t is the output of longitudinal control.

IV. EXPERIMENT AND RESULT

A. Experimental environment

As shown in Figure 8, this paper built an automatic driving platform based on GAC Aion LX. In this platform, power box, host of Integration Navigation System, Nuvo-7160GC IPC, 4G DTU and CAN gateway were arranged. Figure 9 showed the test site which was a circular parking lot with two lanes.



Figure 8. Automatic driving platform based on GAC Aion LX



Figure 9. The test site

B. The result of experiment

(1) Path tracking experiment

This paper did the automatic driving experiment in a parking lot. Figure 10a showed the trajectory of tracking, in which the dotted line is the tracking path and the solid line is the original path. The tracking error is 0-13.51cm, and the average error is 5.42cm. In conclusion, the path tracking algorithm in this paper can track with high accuracy, especially when turning.

The accuracy of tracking largely depended on the shortest value of look-ahead distance l_d . l_d shouldn't be too small, for which it might cause the oscillations of vehicle motion although small l_d would make tracking more accurate. However, l_d shouldn't be too large, for which the vehicle might have a tendency to understeer although large look-ahead distance meant smooth tracking[10].

This paper chose 2m as the shortest look-ahead distance l_0 , as a result of low speed driving in the park.

(2) Lane changing experiment

Based on the high accuracy of tracking, Figure 10b showed the result of lane change. The two different lane change path represented two different position where the obstacles existed. The result implicated that the vehicle was able to change lane safely and smoothly when necessary.

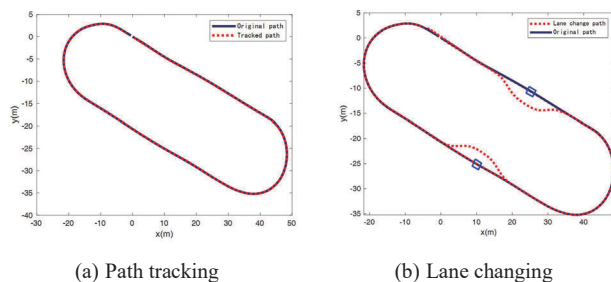


Figure 10. The results of path tracking and lane changing

V. CONCLUSION

In this paper, we proposed a decision-making system including local path planner and control method used in the park with high accuracy and safety. However, its practicability was limited by the scene. In the future, more scenes should be considered, and universally applicable decision-making system should be studied.

REFERENCES

- [1] C. Urmson, J. Anhalt, D. Bagnell, et al, "Autonomous driving in urban environments: Boss and the Urban Challenge," *Journal of Field Robotics*, vol. 25, no. 8, pp. 425-466, Aug, 2008.
- [2] C. R. Baker, and J. M. Dolan, "Traffic Interaction in the Urban Challenge: Putting Boss on its Best Behavior," *2008 IEEE/Rsj International Conference on Robots and Intelligent Systems*, vols 1-3, pp. 1752-1758, Oct, 2008.
- [3] J. Leonard, J. How, S. Teller, et al, "A Perception-Driven Autonomous Urban Vehicle," *Journal of Field Robotics*, vol. 25, no. 10, pp. 727-774, Oct, 2008.
- [4] M. Montemerlo, J. Becker, S. Bhat, et al, "Junior: The Stanford entry in the Urban Challenge," *Journal of Field Robotics*, vol. 25, no. 9, pp. 569-597, Sep, 2008.
- [5] Kammel S, ren, Ziegler J, et al. "Team AnnieWAY's Autonomous System for the 2007 DARPA Urban Challenge," *Journal of Field Robotics*, vol. 25, no. 9, pp. 615-639, Aug, 2008.
- [6] I. Miller, M. Campbell, D. Huttenlocher, et al, "Team Cornell's Skynet: Robust perception and planning in an urban environment," *Journal of Field Robotics*, vol. 25, no. 8, pp. 493-527, Aug, 2008.
- [7] S. H. Bae, S. H. Joo, J. W. Pyo, et al, "Finite State Machine based Vehicle System for Autonomous Driving in Urban Environments," in *20th International Conference on Control, Automation and Systems (Iccas)*, pp. 1181-1186, Oct, 2020.
- [8] Q. J. Cui, R. J. Ding, C. F. Wei, and B. Zhou, "A hierarchical framework of emergency collision avoidance amid surrounding vehicles in highway driving," *Control Engineering Practice*, vol. 109, pp. 104751, Apr, 2021.
- [9] Choi, J. W., Curry, R., and Elkaim, G. "Path Planning Based on Bézier Curve for Autonomous Ground Vehicles". in *World Congress on Engineering and Computer Science, (WCECS)*, pp. 158-166, Oct, 2008.
- [10] Long, H., Yashiro, H., Nejad, H., et al, "Bézier curve based path planning for autonomous vehicle in urban environment". in *2010 IEEE intelligent vehicles symposium(IV)*, pp. 1036-1042, Jun, 2010.
- [11] N. V. Dinh, Y. G. Ha, and G. W. Kim, "A Universal Control System for Self-Driving Car Towards Urban Challenges," in *2020 IEEE International Conference on Big Data and Smart Computing (Bigcomp)*, pp. 452-454, Apr, 2020.